Powering Digital Mega-loads: Barriers and Solutions to Accelerate Grid Efficiency, Investment and GDP Growth in the AI Era

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Abstract – This paper examines how rapid growth in data centre demand interacts with electricity systems that are already facing capacity constraints, decarbonisation pressures and heightened political scrutiny. Focusing on data centres as large-load customers, we distinguish them from both small loads and generation assets to highlight the relevant issues accordingly. Drawing on international literature and case studies from Europe and North America, we identify five systemic barriers to timely and efficient data centre integration: long times to connect; cost uncertainty for both developers and utilities; contested grid cost allocation; capacity constraints; and local political and community opposition. For each barrier, we evaluate emerging regulatory and contractual responses, including deep versus shallow charging regimes, upfront fees and contractual safeguards, flexible and non-firm connections, bespoke large-load tariffs, bring-your-own-generation (BYOG) requirements, location incentives, and socio-environmental and economic-impact criteria. We show that these instruments operate through a limited set of underlying channels and that many can address multiple barriers simultaneously. We argue that fairness concepts in connection policy may need to evolve from strict procedural equality towards outcomes that also reflect net-zero, reliability and economic-development objectives. The paper concludes by outlining how regulators, system operators, planners and data centre developers can use the proposed framework as a diagnostic and design tool, and highlights priorities for future empirical research on policy effectiveness, load-flexibility behaviour and long-term community and system impacts.

Keywords – large load connection, data centre, grid connection policy, energy regulation, connection charges, interconnection, access, shallow or deep connection charges, utilities

I. INTRODUCTION

Energy regulators and system operators globally are confronting the consequences of rapid growth in data centre demand on networks already facing capacity constraints and decarbonisation pressures. Long connection queues, disputes over reinforcement cost allocation, and the growing political sensitivity of new large-load projects have exposed structural weaknesses in existing connection regimes. Governments, regulators and utilities have begun to respond, but policy frameworks remain fragmented and underdeveloped. This paper identifies five recurrent barriers to efficient data centre integration and evaluates policy responses observed across multiple jurisdictions.

A central distinction for this analysis is between load and generation connections. Load connections refer to demand-side connections to the grid, such as households, office buildings and data centres. Meanwhile, generation connections refer to supply-side connections to the grid, mainly power plants. While both have recently undergone regulatory reform in several jurisdictions, this paper focuses on demand-side connection policies, particularly those regulating large-load customers such as demand centres. The economic and system implications of large-load connections differ fundamentally from those of generation projects, particularly in relation to cost allocation, network reinforcement, and system risk.

We further distinguish between small and large loads. Small loads refer to users that consume little energy and individually have no meaningful impact on the grid, such as households or small businesses. Conversely, large loads refer to users that consume large amounts of energy, such as industrial complexes and data centres. Large load users have a big impact on the grid, sometimes requiring significant network upgrades to be connected. Due to this larger impact, large loads are inherently riskier to the grid and costlier. Our article therefore focuses on large load policies, and the connection of data centres to electricity grids.

Within the category of data centres, important heterogeneity exists. Cloud centres typically operate stable workloads; AI training centres often run at very high utilisation for sustained periods; inference workloads fluctuate with user demand; and crypto-mining operations are highly price-sensitive and capable of rapid ramping. These behaviours translate into different risks for the network, particularly regarding predictability, peak demand, and flexibility potential. Their specialised infrastructure also translates into different levels of stranded-asset risks, with AI and crypto-mining centres presenting higher costs for repurposing and larger risk of obsolescence. Some data centres may contain a mix of these services, but generally all data centres will fall within these three high-level categories. While we acknowledge their differences, the issues and solutions presented in this paper can apply, to a higher or lower degree, to all three types of data centres.

Regarding energy networks, we also distinguish between transmission and distribution networks. Transmission networks move bulk power over long distances at high voltages, while distribution networks deliver electricity at lower voltages to final consumers, such as homes, businesses, and most industrial consumers. For electricity to move from generators to final customers, it usually moves through both network levels, although data centres may connect directly to the transmission network. This means that sometimes the energy bottleneck for large load connections to the grid can come from one level or both. Unless otherwise specified, we use "energy networks" in this paper as a generic term which may refer to the individual network level that the data centre is connecting to or the set of distribution and transmission networks.

Energy generation also requires some key distinctions in our paper: renewable and non-renewable energy sources; and dispatchable and non-dispatchable generation. Renewable sources such as wind and solar are low-emission but are variable, and at times produce surplus electricity that would otherwise be curtailed. Non-renewable sources, such as fossil-fuel plants, can provide reliability but conflict with net-zero objectives.

Dispatchable generation can increase or decrease output on command, providing stable and controllable power. Non-dispatchable resources depend on variable conditions and cannot be directed to meet instantaneous demand. Renewable sources such as wind and solar energy are heavily weather-dependent and therefore are generally non-dispatchable. Meanwhile, most non-renewable sources such as coal and gas are dispatchable. This creates a natural trade-off between cleanliness and reliability on energy generation. Further, adding batteries can, to some degree, support both clean and reliability objectives, although batteries add further complexity in terms of commercials, policies and engineering.

This paper analyses five principal barriers we identified which hinder the efficient implementation of data centres in places that would otherwise be ready to receive them. We acknowledge, however, that other barriers may exist which may or may not be addressed by the solutions we provide. It is also important to note that the prevalence of these barriers varies between countries and regions, and our goal is to present an initial assessment of the current practices and conditions internationally, providing a framework against which local ones can be examined.

While the existing literature provides valuable insights into individual aspects of large-load integration (such as tariff design, queue management, flexibility provision, or cost allocation) these issues are often analysed in isolation. This paper contributes by presenting an integrated framework that treats these challenges as interconnected manifestations of underlying structural constraints in electricity systems. We argue that many policy instruments operate through a limited set of economic channels, and therefore reforms should be designed as coherent packages rather than isolated interventions. Mapping barriers and solutions onto shared structural drivers, we provide a diagnostic tool for regulators and policymakers confronting rapid growth in large-load demand.

In section II, we examine the current literature on issues faced by data centres, utility companies and utility regulators. Section III explores barriers we have identified and proposes solutions, based on practices observed worldwide. Section IV identifies common themes between the barriers and solutions, identifying common issues between the barriers, how one barrier impacts another, and how the solutions presented can address multiple barriers at once. Finally, in section V we present recommendations for further research and conclusions from our analysis.

II. LITERATURE REVIEW

The literature on data centre integration into electricity systems has expanded rapidly in recent years, reflecting the acceleration of AI-driven investment and its implications for energy demand. Existing work spans macroeconomic assessments of digital infrastructure investment, analyses of stranded-asset risk, studies of connection charging and cost allocation, and examinations of flexibility, queue management and tariff design for large loads. This section reviews the strands most relevant to large-load connection policy.

The Deutsche Bank Research Institute published a report in 2025[1] which identifies AI-related capital expenditure (capex) in data centres as a significant contributor to recent US GDP growth. The report claims that technology-driven investment has played a stabilising macroeconomic role and is likely to remain an important growth channel in the near term. This framing positions data centre development as both an infrastructure issue and a strategic economic priority. For regulators with mandates extending to economic growth and competitiveness, such macroeconomic considerations may influence the willingness to accommodate large-load projects despite associated system risks.

On the other hand, Arun[40] presents stranded-asset risk as a systemic vulnerability of the current AI and data centre investment boom. The paper argues that rapid capital expenditure, accelerated GPU depreciation, and uncertain revenue streams from AI inference services may generate cascading financial stress across tenants, data centre operators, and lenders. Highly specialised AI facilities are particularly exposed due to limited repurposing options and rapid technological obsolescence. Crucially, this dynamic may extend beyond digital assets: grid reinforcements, generation capacity and transmission investments undertaken to serve large data centre loads could also become stranded. This financial risk perspective is central to evaluating connection safeguards, contract design and cost-allocation mechanisms.

The allocation of connection costs has also been shaped by European regulatory reforms. A 2019 report[2] by the UK Department of Business, Energy and Industrial Strategy (BEIS) summarised European Commission directives encouraging more cost-reflective connection charging, including the recovery of customer-specific reinforcement costs and differentiated treatment of small users. These reforms reflected growing demand from large-load customers and clarified the distinction between shallow and deep charging regimes. Norway subsequently implemented a deep-charging framework for larger connections[26], providing an important case study for the cost-allocation mechanisms analysed in Section III.

Takci et al highlight the increasing need for flexibility in power systems[3], and argue that data centres, as large but technologically controllable loads, can function as flexibility assets. Through workload shifting, utilisation of backup and storage systems, and participation in ancillary-service markets, data centres can provide demand response, frequency support and congestion relief. Empirical examples from major operators suggest that such capabilities have already been deployed in certain jurisdictions. This literature reframes large-load connections as more than mere sources of demand pressure; they can also be seen as potential contributors to system balancing and decarbonisation, contingent on appropriate regulatory incentives.

Cremona and Czyzak[4] argue that grid availability and planning have become decisive determinants of data centre siting in Europe. They document prolonged connection delays in traditional markets such as Frankfurt, London, and Dublin, contrasted with jurisdictions adopting more anticipatory grid investment strategies. The report links AI

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infrastructure expansion to grid readiness, emphasising phased or non-firm connection agreements, flexible load management, and siting of facilities near renewable generation or available capacity as critical measures for enabling growth while supporting net-zero objectives.

One of the constraints to the establishment of data centres in the UK is the long queue to connect to the grid. Ostler et al[6] document severe backlogs and criticise the traditional “first come, first served” (FCFS) allocation rule for delaying projects with high system value. They propose milestone-based accountability, improved transparency and prioritisation mechanisms to accelerate integration of low-carbon and flexible resources. Although their analysis centres on generation, the institutional insights regarding queue design and incentive compatibility are directly relevant to large-load connections.

A 2025 report for the Department for Energy Security and Net Zero (DESNZ)[9] aimed to measure the impact that growth in data centres would have on energy *consumption* (not wider issues such as grid connections) in the UK. The study develops a methodology to compare electricity consumption in digital and physical service delivery chains to estimate the net energy impact of digitalisation. It finds that while data centres increase overall electricity demand, they can displace more energy-intensive physical processes, generating a net impact that may be beneficial in terms of reducing overall energy demand. For instance, it demonstrates that digital delivery of books and translations consumes a fraction of the energy used by their physical equivalents, while video streaming and Blu-ray viewing are broadly comparable when manufacturing occurs in the UK.

The report also distinguishes between centralised and decentralised energy use, noting that around a quarter of digital energy demand is concentrated in data centres, compared with much higher centralisation in physical manufacturing. Overall, digitalisation is found to improve decarbonisation readiness, since data-driven activities are already fully electrified and therefore well-positioned to align with the UK’s decarbonisation of the power system.

The report shows that the energy efficiency of UK data centres has improved markedly over the past decade, with Power Usage Effectiveness (PUE) values converging toward an industry average of 1.3–1.6. Newer hyperscale facilities achieve even lower ratios due to advances in cooling and server optimisation, supported by a moderate UK climate and industry commitments such as the Carbon Neutral Data Centre Pact. However, the report also recognises that efficiency gains are slowing as technical limits are approached, and that emerging high-intensity computing applications, especially Generative AI, could increase both localised power demand and the complexity of grid connections.

Pollitt et al[5] explored the possibility of a more market-based approach to manage the connection queue in the UK (note that work was funded and supported by one of the UK network companies, UKPN). They focused on the limitations of the current first come first served (FCFS) approach, noting that it does not account for the feasibility, progress, or wider system value of projects seeking connection. The paper explored potential changes to the initial allocation of connection rights, such as establishing auctions or “beauty contest” criteria, and the possibility of introducing a secondary market in which projects could voluntarily trade queue positions.

The authors analysed the connection queue by examining how the current FCFS allocation rule operated in practice, comparing its properties against established results in auction theory, mechanism design, and queuing theory. They used available queue data and internal UKPN documents to map the sequence of steps needed to enter the connection queue. This description was then used to assess whether the FCFS process satisfied conditions such as incentive-compatibility, strategy-proofness, and efficient allocation as defined in the literature.

To evaluate alternatives, they applied formal models from the auction and queuing theory literature directly to the distribution-level connection characteristics, assessing incentive compatibility, truthfulness, and the feasibility of secondary trading given substation and reinforcement constraints.

The paper concluded that the FCFS system is not incentive-compatible and contributes to speculative applications and inefficient queue outcomes. It identified several potential improvements, such as reviewing connection fees, requesting certain connection costs to be paid upfront or certain costly milestones to be met before getting placed in the queue, considering non-price criteria for prioritisation in the connection queue; and enabling voluntary trading of queue positions between customers, subject to restrictions designed to avoid impacts on third parties and to manage risks such as scalping, asymmetric information, and behavioural biases.

After updating its connection policy in April 2025[7], Ofgem noted that the demand queue for connections in the UK increased more than its highest projections[8], driven mostly by data centre projects. Ofgem evaluated that this increase needed to be swiftly addressed and planned a series of actions to determine which projects to prioritise and possible ways to speed up the queue without hurting its regulatory duties.

In the USA, data centres are posed as one of the main drivers for future increases in energy demand. A 2024 report by the North American Electric Reliability Corporation (NERC) noted[10] that demand growth in Canada and the USA was higher than at any point in the previous twenty years, and that projections of energy growth over the next ten years were still steadily rising. According to the report, in most areas of North America the projected demand growth is mostly driven by data centres and other large commercial and industrial loads, such as hydrogen fuel plants, smelters, manufacturing centres and electric vehicle infrastructure.

Satchwell et al[11] built on this assessment by identifying practices in tariff design that could help governments and regulators to achieve their desired policy and financial outcomes. The paper cites policies on:

- Eligibility and applicability, such as minimum load requirements and monthly demand charges;
- Contract size, pertaining to contracted capacity and load factors;
- Contract duration and exit fee;
- Generation requirements, mostly focused on renewable sources;

- Other practices such as marginal pricing and economic development payments.

The authors assert that these elements can help governments and regulators to reach their goals regarding ensuring resource adequacy and affordability and promoting reductions in pollutant emissions. At the same time, they argue that some of these policies may also help attract new data centre projects, which is desired by some local governments to promote economic growth in their region.

III. Barriers and Solutions

The identified barriers and our evaluated solutions are listed below. While their underlying conditions are common, their manifestation varies across countries and regions.

Before turning to the specific barriers and solutions, we highlight the broader trade-offs regarding electricity connection policies.

Firstly, policies could work in opposing – not complementary – directions. For example, subsidies (from governments or other grid users) that benefit data centre loads could inefficiently increase grid demand for low-value loads at the expense of high-value loads. This reflects the tension between attracting investment and ensuring that new demand generates net system value. While electricity is a standardised product, generation costs (e.g. carbon), and consumption benefits, vary greatly.

Secondly, standardised charges – rather than cost-reflective pricing – tend to dominate. For example, socialising connection charges ("shallow charges") means that the connecting party (i.e. a data centre) does not pay the full cost of the network upgrades, so these are instead borne by other grid users (e.g. residential users).

Thirdly, regulation – not competition – dominates the policy landscape. As a result, procedural rules and legal precedent can outweigh economic efficiency considerations. By extension, process, precedents, laws and rules can dominate economics, growth, efficiency and business logic.

Fourthly, fairness – not efficiency – dominates regulatory rules, as noted by Biggar[36]. Thus, even if a data centre offered higher payments for connection and use of the electricity system, an earlier applicant could challenge preferential treatment on fairness grounds. Relatedly, Khezr[37] suggests auctioning capacity to connecting parties could be more efficient than uniform access pricing (and the related connection queues).

Using the UK (specifically Great Britain, GB) as an example, connection policy has historically treated fairness as synonymous with equal treatment and non-discrimination, which in practice has meant allocating queue positions on a FCFS basis. These fairness duties were established long before regulators were given explicit legal obligations to support net zero and economic growth. Ofgem only recently received a statutory net zero duty, along with the powers to enforce it, in the Energy Act 2023[38]. Likewise, the duty to support GDP growth did not apply to Ofgem until 2024[39]. This sequencing means that Ofgem and regulated network companies are expected to prioritise net-zero and growth-positive projects, while still avoiding "undue discrimination" between customers, making it legally and conceptually difficult to favour projects that better serve wider policy goals.

These tensions help explain why legacy FCFS around the world are being increasingly questioned in the literature. Fairness as strict procedural equality becomes hard to justify when the underlying system is increasingly constrained and when delays impose social and economic costs. In this context, fairness assessments would also consider outcomes, such as reduction of system risk, decarbonisation support, or economic and social benefits. Different jurisdictions have responded differently, reflecting variations in legal frameworks and market design.

However, some regions rely more on market-based approaches than others. For example, in the US, competitive processes are reflected in bespoke bilateral contracts. By contrast, most jurisdictions, including the UK, have not generally used auctions to allocate connection capacity, although competitive mechanisms are used elsewhere in the energy sector to allocate scarce rights, suggesting a possible future role if designed carefully. We explore how an auction mechanism would interact with our proposed solutions in section IV.

The solutions we present in this paper sit along this spectrum: from heavily regulated rule-based tools to more flexible market-driven mechanisms.

A. Time to connect

Long connection times typically arise from queue congestion or legal challenges by local opposition.

In many instances, local energy grids do not have sufficient capacity to accommodate all connection requests, leading to network assessments and improvements being necessary before new projects are connected. Prolonged waiting times may prompt developers to relocate or abandon projects.

In a special report on energy and AI[17], the International Energy Agency (IEA) presents estimates of average time to connect to the grid in some countries. In the UK, the estimates ranged from 5 to 7 years. Meanwhile, in the US, the average time was between 1-3 years, with places like North Virginia presenting an average waiting time of up to 7 years.

Proposed solutions

a) *Requiring data centres to pay upfront*

Requiring data centres to bear upfront costs could help minimise speculative data centre connection requests. Speculative requests for connections are either not backed by a finalised project but serve as a way to hold a place in queue as the project is developed, or an existing project that applies for connections in multiple locations with the intent of building in only the one that gives them the lowest cost[24]. Speculative applications can inflate connection queues and drive away value-adding projects. Advance charges may create a reasonable and simple barrier and reduce the number of speculative requests although there could be downsides (lack of visibility and rent extraction by monopolies).

As part of its new regulator-approved Schedule Data Center Tariff (DCT), the power company AEP Ohio, in the USA, adopted such an approach[25]. For a data centre project to be allowed to remain in the connection queue, it must submit a study request and pay a fee that can go from US$ 10,000 to US$ 100,000, depending on the size of the capacity request. The fee must be paid within 45 days, or the applicant forfeits its queue position. Following the load study,

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applicants must also commit to reimbursing 100% of buildout costs if the project is cancelled or delayed by more than 12 months. These provisions discourage speculative applications and protect existing consumers from stranded upgrade costs.

Additionally, charges could reflect the type of data centre, given their expected benefit (or risk) to the local economy and other grid users. For example, Crypto-mining loads may pose higher stranding risk. This could help attract projects that offer more benefits (net zero, economic growth, reducing grid charges for others, etc) at the expense of loads that bring higher risks or costs.

b) *Flexible Connections*

As suggested in the paper by Takci et al[3], data centres can be a source of flexibility for the energy grid in the UK. Knittel et al make a similar argument[19] for the USA. The papers mention that data centres can contribute to grid flexibility in the following ways:

- **Workload shifting:** data centres can reschedule computational tasks from peak to off-peak periods, reducing demand during system stress and increasing it when spare capacity exists. Shifting load away from peak periods reduces reliance on peaking generation and allows greater utilisation of baseload and renewable resources, lowering system costs and supporting reliability. In addition, workloads can also be shifted to periods of high renewable output or low market prices, improving the utilisation of zero-carbon generation and reducing renewable curtailment.
- **Leveraging delay-tolerant workloads**: many workloads, such as batch processing, analytics, or backups, can tolerate delays without affecting user experience. These tasks can be rescheduled strategically to adjust power demand in line with grid conditions.
- **Participation in demand-side response schemes**: through smart energy management algorithms, data centres can reduce or increase their demand in response to system needs, alleviating congestion, reducing peak loads and supporting real-time balancing requirements.
- **Spatial workload migration**: computational tasks can be moved between geographically distributed data centres, allowing operators to align load with regions where renewable generation is abundant or where the local grid requires additional flexibility.
- **Bidirectional power flow**: modern data centres have the capability not only to draw electricity but also to export surplus power back to the grid, enabling them to function as either flexible demand or auxiliary supply depending on system conditions.
- **Use of UPS systems for ancillary services and as virtual energy storage**: the high-capacity UPS batteries installed for reliability purposes can provide fast frequency response and other ancillary services, contributing rapid stabilising energy to the grid without the need for dedicated storage infrastructure. Data centres can also use their battery capacity as a virtual energy storage resource, displacing the need for standalone grid batteries and contributing to reserve requirements.
- **Thermal energy storage and thermal inertia**: cooling systems and thermal storage tanks can shift cooling demand by storing excess cooling when renewable energy is plentiful or during off-peak periods and discharging it during peaks. This reduces grid stress and supports demand-side flexibility.
- **Reducing transmission losses and congestion**: geographically distributed data centres can provide decentralised flexibility, alleviating power flows on constrained transmission corridors, reducing losses and helping maintain system stability.

The authors recognise, however, that government incentives and clear legislation might be needed to push data centres towards reaching their potential on this matter. Takci et al mention that data centre operators can be cautious about participating in demand response or ancillary service markets due to Quality of Service (QoS) and Service Level Agreement (SLA) requirements. Lack of legislation permitting data centre participation in energy markets or ancillary services can also be a deterrent to the implementation of certain features in some countries.

To ensure data centres deliver on this potential when the necessary regulatory and grid conditions are met, their connection contracts can include requirements on fulfilment of certain capabilities. Companies that did not want to commit to these duties could be required to only connect on a flexible basis, such as under interruptible provision. However, both Data Centres and Supplier companies in the USA have opposed a proposed imposition of flexible connections by the PJM[20] citing legality, fairness and grid reliability.

c) *Network company incentives*

The UK regulator, Ofgem, uses many different types of financial incentive to encourage network companies to provide better connection services.

Connection incentives change between price controls (RIIO-2 differs from RIIO-1) and between sectors (distribution companies have different incentives than transmission companies).

For example, distribution companies have “connection engagement” and “Time to Connect” incentives across both RIIO-1 and RIIO-2. By contrast, the transmission companies had an incentive regarding “Offers of timely connection” during RIIO-1 which was replaced by two incentives for the RIIO-2 period: “Timely connections” and “Quality of connections satisfaction survey”. Notably, the “Timely connections” incentive was designed to be penalty-only and Ofgem said that a material penalty would apply only if there was significant deterioration or a systematic failing.

Further, Ofgem applies guaranteed standards, although currently these apply only to distribution not transmission networks[43].

B. Cost uncertainty

The two primary cost uncertainties are construction and connection.

Regarding construction uncertainty, there are many costs associated with building a data centre. In addition to regulatory and connection costs, companies face

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construction, material and labour costs. According to the Turner & Townsend Data Centre Construction Index Report of 2025[18], most stakeholders expect data centre construction costs to increase by more than 6% in 2026.

Crucially construction costs vary greatly by region to such an extent that attracting data centre loads (by offering favourable connection terms, grid subsidies or tax breaks) may be ineffective. Turner & Townsend's report mentions that due to Tokyo being one of the most expensive places to build data centres in the world, companies are deciding to establish their facilities in other regions of the country, such as Osaka.

While location can be a major source for cost differences, the approach to connection charges by the local energy company can also have a large influence on cost uncertainty. In regions where the data centres are required to pay for grid reinforcements ("deep connection charges"), the connection costs are likely to be higher than other regions.

In terms of uncertainties to the grid, data centres bring a large risk to the network given their characteristic as large energy users. The possibility of these facilities becoming stranded assets require network companies to consider steps to prevent this risk from affecting their network upgrade investment recovery, preferably without burdening other consumers. This issue is one that affects both cost uncertainty from a utility company point of view and cost allocation from the side of local communities.

Proposed Solutions

a) *Safeguards in connection contracts*

Data centre tariffs and special contracts could include safeguards[28] regarding contract length, minimum demand, minimum load factors, power factor range, investment bonds, cost reflectivity, planned load-shedding, unplanned interruptible load value, and stated times or durations for interruptions.

The utility company in Ohio, AEP, established minimum demand charges within its Data Center Tariff[25], where the monthly billing payment will be a percentage of the contract capacity, or 85% of the customer's highest previously established monthly billing demand during the previous 11 months, whichever is higher. The power company also implemented a minimum contract length that equals the Load Ramp Period plus eight years, to a maximum of 12 years.

These types of measures reduce the risk that data centres pose for local networks, while also providing more accurate load profiles. This reduces cost uncertainties for utility companies, which in turn can provide a lower variability in prices, increasing predictability for data centres' cost projections.

b) *Permission for bespoke contracts*

Data centres could negotiate bespoke contracts with network owners and/or operators, under regulatory approval. These contracts could be used to get the data centres to contribute to the local network to the best of its ability and could result in lower costs for the energy company and lower risks for other grid users. Upon the bilateral arrangement, requesting companies could be prioritised in the connection queue, leading to a more efficient management of the queue as well.

In 2025, Indiana Michigan Power (I&M) and Google filed a special contract[31] with the Indiana Utility Regulatory Commission (IURC) to support the tech company's new Fort Wayne data centre. The main mechanisms introduced in the contract were a Clean Capacity Arrangement (CCA), which mostly benefits I&M by helping it meet its annual capacity obligations within PJM, and demand response participation, which should benefit other grid users.

Under these demand response arrangements, Google agreed to reduce a set amount of its electricity use when requested by I&M during times of high demand, within limits in terms of how often, how long, and how many times per day these interruptions can occur. Additionally, Google has agreed to follow PJM's emergency demand response rules, which can protect small consumers during certain emergency events from having problems with energy supply.

Through these agreements, Google's participation reduces I&M's peak load and associated capacity and transmission costs, with resulting credits recovered through I&M's existing cost-recovery mechanisms.

Bespoke contracts could not only lead to better outcomes for the energy system but also reduce cost uncertainty for the data companies. This could be accomplished by pre-emptively negotiating connection conditions, so that bespoke conditions are reflected in grid plans and connection studies.

By contrast, whilst counter-productive and counter-intuitive, it is entirely possible under a FCFS policy that a data centre that would easily offer substantial benefits to other grid users could be delayed (or lose out entirely) in favour of another connecting party that increases risks and costs for other grid users.

C. Grid cost allocation

One of the main concerns raised by local communities is how the costs and risks are distributed among grid users.

Data centre developers argue that it would be unfair for them to have to cover all the costs for the network upgrade, given that upgrades should ultimately benefit the local grid. By contrast, consumers argue that they should not pay without there being an immediate benefit and that network upgrades are only necessary because of the data centre.

In the USA, several states have a shallow connection charge policy. It has been estimated that customers from Illinois, Maryland, New Jersey, Ohio, Pennsylvania, Virginia and West Virginia, collectively paid over US$ 4.3 billion[27] in 2024 for grid investments that were only necessary to accommodate new data centres. This figure demonstrates how cost allocation can affect local communities where companies decide to establish their data centres.

While shallow connection charges can help attract new data centre projects, they can have a negative impact in the local community's cost of living, which could then fuel counter-productive opposition.

In 2022, Ofgem moved towards "fully shallow" demand-side connection charges[13], such that "100% of reinforcement costs" are funded by other customers. In 2025 the Northern Irish Department for the Economy moved in a similar direction[41]. However, if demand is going **up**, it may appear odd that the price is going **down**.

Nonetheless, it is not a free lunch for connecting parties – the benefits are capped: Ofgem added a high-cost cap of

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£1720/kVA for demand connections[42], while the Utility Regulator in Northern Ireland was less generous, proposing a high-cost cap of £1000/kVA for demand connections.

Going in the opposite direction, Norway established in their latest policy update that a deep connection charge would be used for projects with capacity above 1 MW[26]. Under this framework, data centres that have a required capacity above the specified level must cover the costs of any network upgrade that is necessary for their connection to the grid for 1st (same capacity level) and 2nd order (one level above) connection costs.

Proposed Solutions

a) *Negotiate or mandate grid connection costs*

Sometimes, the establishment of a data centre, especially a large one, may require grid improvements or reinforcements. Under shallow connection charges, this cost is likely to be distributed among all consumers, and existing customers will end up indirectly subsidising the new data centre connections, even if those data centre developers are willing to pay for grid improvements and reinforcements in exchange for an earlier connection.

There are several ways that deep connection charges can be implemented. They can be mandated by regulators, such as in Norway, or other policy makers. Alternatively, energy companies can make their own connection guidelines establishing deep charges, with the support or authorisation from the regulator.

Alternatively, data centre developers could negotiate with network companies and/or seek derogations from standard rules, in exchange for prioritisation in the connection queue. This option would allow negotiations to reflect date-specific and location-specific factors.

b) *Differentiated rates for data centres*

Data centres could pay bespoke (e.g. higher) use of system costs. According to a Berkeley report from 2024, data centre loads in the USA are not necessarily flat and may fluctuate over time[30], with different types of servers and tasks leading to different levels of variation in load usage over time. AI data centres, for example, were estimated to operate on average at a relatively stable 80% capacity when performing model training, while operating with a much larger variability, averaging 40% total capacity usage, when doing inference tasks.

To compensate for the higher risk to the grid that this variability represents, data centres could voluntarily offer higher grid-use charges, or it could be mandated by regulators or policymakers. Alternatively, these higher rates could be negotiated between respective parties, subject to agreement with the regulator or policy maker. Again, this option could allow negotiations to reflect date-specific and location-specific factors.

In the USA, Idaho Power Company[29] applies a special tariff to customers whose metered usage exceeds 2 MWh for at least three billing periods and who have a minimum demand of 1 MW. This allows the company to differentiate between smaller, less risky consumers and bigger, riskier ones.

Rate differentiation works similarly to deep connection charges in terms of allocating costs, both monetary and in terms of risks, to the originator (data centres) rather than the larger customer base, which is likely to comprise mostly, if not completely, of small energy users.

D. Capacity constraints

In the UK, the Energy Networks Association (ENA)[14] reported that the queue for demand connections increased to 125 GW as of June 2025. This is approximately 18.6% higher than the total generation capacity of the UK grid, according to the most recent data from the Digest of UK Energy Statistics (DUKES)[15], which encompasses the year of 2024. Total installed capacity and total de-rated generation capacity in the UK are 105.4 GW and 71.7 GW, respectively.

The situation becomes even more difficult when accounting for capacity differences between transmission and distribution. The queue demands were 97 GW from transmission and 29 GW from distribution. In contrast, total installed capacity is 66.8 GW for transmission and 38.6 GW for distribution.

Proposed Solutions

a) *Bring Your Own Generation (BYOG)*

Initially, it may seem that the most straightforward solution to capacity constraints is for data centres to bring their own generation or enter into Power Purchase Agreements (PPAs) with generators.

However, the American network operator, PJM, reveals several complications in its May 2025 publication, showing 8 options for large load co-location, identifying unique grid impacts of each[21].

The Irish regulator, CRU, applies size-specific rules to data centres[12].

Small data centres (<1 MVA) follow the standard connection process. No additional policy requirements apply, but the System Operator (SO) must assess whether the connection is in a constrained or unconstrained location.

Medium-size data centres (≥1 MVA and <10 MVA) must provide an "autoproducer unit"[44] with de-rated capacity equal to 100% of the site's maximum import capacity (MIC), operational for the duration of the connection and participating in the wholesale electricity market. They must also meet at least 80% of annual demand with additional renewable electricity generated in Ireland within six years of energisation. Where compliant, no separate Mandatory Demand Curtailment (MDC) requirement applies.

Large data centres (≥10 MVA) are subject to the same renewable requirement but must instead provide dispatchable onsite or proximate generation and/or storage, on a de-rated basis, equal to their MIC. This capacity must be separately connected and metered, participate in the wholesale electricity market, and be delivered before the site may operate or ramp to full MIC. Where compliant, no separate MDC requirement applies.

Specifically, CRU does **not** allow a data centre to meet its capacity obligation using remote generation elsewhere in Ireland. However, it **does** allow remote renewable generation to count toward the 80% renewable requirement.

Over time, these arrangements are expected to facilitate timely data centre connections without breaching system security standards, provide a revenue stream that can offset investment costs, and encourage the deployment of future-proofed low- or zero-emission technologies, supporting both

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operational reliability and the long-term decarbonisation of the electricity system.

Overall, the CRU provides a model for regulation-driven BYOG, where data centres contribute to net zero targets. However, emissions could still increase rather than decrease on a marginal and peak-period basis because the 80% renewable requirement: 1) permits 20% non-renewable generation, 2) does not require temporal matching, 3) does not require locational matching, and 4) is softened by a 6-year glide path.

As of July 2025, the Irish government is working on a Private Wires Policy[22] which would facilitate PPAs between data centres and renewable energy generators. This step is considered highly beneficial towards decreasing time to connect, as well as reducing reliance on network supply.

b) *Location incentives*

Complementing BYOG initiatives, governments and regulators can also provide incentives for data centres to establish themselves in locations where there is excess, constrained off, renewable energy. This would focus on areas where due to some factor such as low demand or constraints, the energy generated by these renewable sources would be wasted (curtailed or constrained off).

Additionally, data centres could be incentivised to settle in areas of high renewable energy potential. Stimulating data centres to be set up in such areas and building alongside them renewable generators with sufficient de-rated capacity in line with their MIC would not only help speed up the connection queue and circumvent capacity constraints but also promote renewable energy and reduce risks for the larger network.

E. Local opposition

The establishment of data centres can also be met with political resistance from the local population and politicians. This creates an additional constraint to the installation of these facilities, possibly leading to delays or abandoned projects.

The Data Center Watch organisation reported[16] that, in the USA, US$ 64 billion worth of investments in data centres have been blocked or delayed by local opposition between May 2024 and March 2025. According to the report, US$ 18 billion worth of projects has been completely blocked, while US$ 46 billion have been delayed, due to resistance from residents and activist groups.

The report states that the main concerns from local communities go beyond resource allocation, and encompass issues such as visual impact, noise pollution, and effects on property values, as well as negative environmental impacts locally. In Prince William, Virginia, the local community also raised concerns over potential damage to nearby historic sites, and a US$ 24.7 billion project remained, as of March 2025, still delayed due to ongoing legal challenges by local groups.

While opposing politicians usually mirror the concerns of the local opposition groups, some also cite proposed tax incentives as a reason to dispute data centre projects.

This local opposition both reflects and adds to other constraints mentioned in this section, which are more economic in nature. While economic measures can address some concerns, such as strains on energy provision and cost allocation issues, others might need more nuanced solutions, which can add complexity and uncertainty to data centre projects.

Proposed Solutions

a) *Socio-environmental criteria for authorisation and or prioritisation of projects*

Data centres compliant with net zero initiatives and with the best-performing Economic Impact Assessment (EIA) or highest in-country-value could be prioritised. Compliance with net zero could reflect energy efficiency and emission impact, which would likely be lower for those with renewable energy generators. Additionally, such criteria could be included as requirements for data centres to be allowed to connect to the grid. BYOG could also be used as a criterion for prioritisation in areas where making it a requirement might be unfeasible due to space or infrastructure constraints.

The 2023 Energy Efficiency Directive published by the European Parliament[35] provides an example of how a data centre energy efficiency assessment can be made. The document requires data centres to communicate their levels of energy consumption, power utilisation, temperature set points, waste heat utilisation, water usage and use of renewable energy.

The goal is to initially use the provided information to set a baseline against which other data centres can be compared and evaluated. Once set, this baseline could also be used to evaluate future data centre projects and establish a minimum efficiency threshold for construction authorisation. In terms of queue management, authorised projects could be prioritised based on expected energy efficiency, incentivising companies to invest low-energy consumption technology, as well as good quality on-site renewable generation.

On economic impact, the projects with the highest in-country benefits, such as supporting domestic supply chains, manufacturing, labour and local communities, could be prioritised. This would benefit local communities and would align with the mandates that some regulators have on supporting GDP growth. EIA-based criteria would also ensure that the expected benefits to GDP growth of allowed projects would go beyond the direct contributions from the data centre profits and the increased profits from the utility providers.

In Indiana, USA, the Indiana Michigan Power (I&M) offers a billing credit[32] to large load users that comply with some conditions. Among them, new customers are required to create at least 20 full time equivalent jobs or invest at least US$ 2 million at the service location. While these are not conditions for connection eligibility or prioritisation in the connection queue, they provide a good example of the type of conditions that could be used for that.

Through its Gate 2 methodology[33], UK's NESO established readiness and strategic alignment criteria for a data centre project to be prioritised in the connection queue. In summary, projects which meet minimum size requirements, have the size boundaries properly set, have secured land rights, and submitted a proper application for planning consent are considered to have met the readiness criteria.

For the strategic alignment, one of the following criteria must be met:

- The project is eligible for relevant protections;

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- The project is aligned to the capacities within the Clean Power 2030 (CP30) Action Plan;
- The project is designated as described in the Project Designation Methodology; or
- The project is not within scope of the CP30 Action Plan and of a specified technology type.

While some projects may be exempt from having to comply with environmental guidelines, as determined by the strategic alignment criteria, NESO's policy showcases how net zero alignment can be used to prioritise projects in the connection queue.

In Ontario, Canada, a bill[35] is being discussed which would allow the Ontario Energy Board to establish requirements on economic development and job creation for large energy users to be approved for connection to the network. This initiative promotes benefits for the local community, while mitigating the number of applicants for network connections, thus reducing the size of the connection queue and the pressure on the local grid.

Having projects that align with environmental goals and that support the local economy beyond the data and utility companies would address some of the main concerns of local communities. These measures would also likely improve the perception of fairness by the general population, reducing local opposition to the establishment of data centres in certain regions.

IV. Common Themes and Systemic Interactions

As we mentioned in the previous section, some barriers may originate from the same underlying issues, which means that solutions that address the issue will likely reduce both barriers at the same time. Whether this is the case in any specific application is highly dependent on the local conditions of the region or country being examined. We present below the most likely overlaps and expected interactions among barriers and proposed solutions.

The main issue that affects all five barriers is the structural limitations in the energy network and the fast-paced growth of demand for connections by data centres. Long connection times often arise because local grids are not fit for the number of data centre requests, which in turn increases the need for network upgrades and heightens uncertainty over both timing and cost. This same need for network reinforcement also fuels disputes about grid cost allocation, as communities question whether they should bear the cost of upgrades required primarily for data centre development. Where these cost-allocation concerns translate into higher bills or visible system pressures, they can reinforce local opposition, creating further delays through political resistance or legal challenges. In this way, bottlenecks in one part of the system tend to amplify constraints in others.

Several proposed solutions therefore interact in mutually reinforcing ways. Measures such as flexible connections, workload shifting, demand-side response, and BYOG can all ease the pressure on grid capacity. By reducing peak demand, supporting balancing requirements, improving utilisation of existing infrastructure and contributing to energy generation, these solutions can help shorten connection queues and reduce the scale of network upgrades required. This in turn can reduce both construction and connection cost uncertainty, while also lowering the perceived burden on local communities, indirectly helping to mitigate opposition.

Other solutions operate by targeting the same underlying issue of speculative or poorly sequenced project requests. Requiring data centres to pay fees upfront discourages speculative queue entries and helps ensure that projects progressing through the queue are backed by firm commitments. Similarly, safeguards in connection contracts, such as minimum demand levels, minimum contract lengths and power-factor requirements, reduce risks for energy providers and improve forecasting. These mechanisms make it easier for network companies to plan investments and reduce the risk of stranded upgrades, thereby lowering cost uncertainty and making the grid more resilient to rapid growth in data centre demand.

Several solutions work through reducing the scale or location-specific impact of demand. BYOG requirements, for example, can offset part or all the data centre's demand on the grid. This reduces pressure on constrained networks and can help accelerate connections in areas where capacity is already tight. Location incentives that direct data centres towards regions with surplus renewable generation or previously underutilised energy resources also help alleviate network constraints while supporting renewable-energy utilisation. Both approaches reduce localised grid stress and can limit the number of upgrades required, thereby diminishing conflicts over cost allocation and reducing grounds for local resistance.

There are also solutions whose interactions arise through cost allocation and fairness considerations. Deep connection charges ensure that data centres bear the cost of necessary network upgrades, preventing existing consumers from indirectly subsidising new connections. Differentiated rates and bespoke contract structures can align ongoing charges with the system risks and benefits associated with each project. These mechanisms can reduce conflict between developers and communities and may also help manage uncertainty for both sides by clarifying cost responsibilities early in the project lifecycle.

Finally, solutions that incorporate socio-environmental criteria for authorisation or prioritisation connect the economic, environmental and social dimensions underlying multiple barriers. Prioritising projects that demonstrate strong energy-efficiency performance, renewable-energy usage, or high economic-impact potential can reduce local resistance by ensuring that the projects most beneficial to communities and national objectives move forward more quickly. Such criteria also support more efficient queue management by reducing the number of applicants and by aligning project selection with broader policy goals, including net-zero commitments and local economic development. Including BYOG as a prioritisation criteria can also add grid capacity benefits to this list.

One way to enforce many of the solutions proposed here without a large increase in regulatory burden is by allowing regulator-supervised auctions of connection rights. In such a system, network companies would be able to auction their available capacity to the highest bidders (these being the companies willing to pay the most for new connections or the ones willing to provide the most safeguards), ensuring that the companies that would be allowed to connect are the ones that have the highest projected economic outcome, likely

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translating into larger GDP impact. By itself, this would already greatly reduce the number of speculative projects asking to connect to the grid, increasing the efficiency of the connection allocation. Coupled with socio-environmental requirements for participation in the auctions and deep connection charges, this solution can address many barriers while promoting GDP growth and requiring little additional regulation.

Overall, many of the solutions proposed address more than one barrier simultaneously because the barriers themselves arise from shared structural issues: insufficient grid capacity, inefficient queue management, uncertainty over costs, and community concerns about fairness and local impact. Understanding these interactions is essential for designing policies that maximise system-wide efficiency and ensure that data centre growth contributes positively to energy security, economic development and environmental goals.

V. Conclusion and Recommendations

This paper has examined five key barriers to the timely and efficient integration of data centres into electricity systems: long times to connect, cost uncertainty, controversial grid cost allocation, capacity constraints, and local opposition. Drawing on international evidence and emerging regulatory practice, we have outlined relevant policy and contractual tools that can mitigate these barriers, and we have shown that many of them operate through common underlying channels. Structural constraints in network capacity, rapid and concentrated growth in large-load demand, and outdated queue management and cost allocation create reinforcing pressures that manifest simultaneously and create these observed barriers.

Regulators, system operators and policymakers can use this framework as a diagnostic and design tool. Rather than treating each barrier in isolation, they can start by mapping local conditions against the five barriers, identifying which structural issues are most pronounced (for example, transmission bottlenecks versus distribution constraints, or speculative queues versus genuine capacity shortfalls). From there, the solutions presented can be combined and sequenced in ways that reflect the local technical, regulatory and political constraints. Our key recommendation is that reforms should be designed as a coherent package: queue management, cost allocation, flexibility incentives and socio-environmental screening work best cohesively, rather than introduced separately.

For regulators and network companies, one practical way to use this work is as a structured checklist when revising large load connections and tariff policies. When assessing a proposed reform, regulators could ask: which of the five barriers is this primarily addressing; which others will it indirectly affect; and what complementary measures are needed to avoid simply shifting the constraint elsewhere in the system? Likewise, case studies such as Norway's deep charging regime, Ireland's BYOG-based connection framework, and US bespoke contracts for large data centre customers can be used as reference points but should not be copied wholesale without adapting to local institutional and system conditions. Where bespoke agreements are permitted, regulators may wish to articulate clear principles on transparency, non-discrimination and consumer protection so that individual deals contribute to, rather than undermine, the wider policy objectives.

For governments and planning authorities, the findings underline the importance of aligning data centre strategy with energy, industrial and regional development policy. Location incentives, socio-environmental criteria and economic-impact requirements can all help steer investment toward areas where data centres can relieve rather than exacerbate network constraints, make better use of surplus or stranded renewable resources, and deliver visible benefits to local communities. However, these instruments should be designed with an explicit awareness of trade-offs: for example, stronger BYOG and net-zero requirements may improve long-term system resilience and decarbonisation outcomes but may also raise short-term costs, discouraging immediate projects. In practice, it may be preferable to provide alternative compliance options, rather than a single prescriptive model, so that companies can more easily comply with the regulation without compromising the benefits to the grid.

For data centre developers and operators, the paper highlights both risks and opportunities. Stricter queue management, deep connection charges and enhanced socio-environmental criteria will likely increase the burden of due diligence and raise the cost of speculative behaviour. At the same time, developers that can credibly demonstrate flexible load management, voluntarily participate in demand-side response, invest in local generation or storage, and provide consistent local economic benefits may be able to secure faster connections, more favourable contractual terms, and greater political support. The recommendations therefore point toward a more partnership-based model in which large users are not merely grid customers but system resources that actively contribute to reliability and decarbonisation.

The analysis also exposes several limitations and gaps in the current evidence base, which shape our recommendations for future research. First, much of the literature and many of the case studies remain descriptive or conceptual. There is a need for quantitative evaluations of the policy instruments using robust counterfactual methods. Future work could, for example, compare regions that have implemented different combinations of these tools, examining impacts on connection times, grid investment costs, consumer tariffs, and the pace and composition of data centre deployment.

Second, some of the papers we analysed have shown the importance of load profiles and flexibility potential, but empirical evidence on how different types of data centres (data centres, AI facilities, crypto-mining and mixed-use facilities) behave under different tariff and regulatory regimes remains limited. Further research could examine these load profiles in more detail, quantify their contribution to system flexibility under realistic operational constraints, and assess the risks to the grid that each type poses.

Additionally, we note there is relatively little systematic evidence on how different types of data centres affect community welfare and long-term regional development. Future research could focus on the impact of each type of data centre and quantitative tariff-impact assessments under different regulatory models. This would help policymakers to better estimate the advantages of attracting data centres and regulators to calibrate contract safeguards, flexibility obligations and incentive schemes more precisely.

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Finally, future work could explore the impact on net zero goals of data centre integration under different energy transition pathways. Scenario analyses that jointly model data centre growth, renewable deployment, network reinforcement, and connection policies would allow decision-makers to understand long-term trade-offs and path dependencies. Such analyses could also incorporate uncertainty around technology costs, AI demand growth and regulatory responses, helping to evaluate the robustness of policy packages.

In summary, the rapid growth of data centre demand is not inherently incompatible with grid reliability, affordability or net-zero objectives, but it does require a deliberate and integrated policy response. By viewing the barriers identified in this paper as interconnected results of underlying structural issues, regulators and policymakers can design more coherent interventions that not only solve issues brought up by data centre connection demands but can also leverage these facilities to support grid reliability and environmental goals. The framework and examples presented here are intended as a starting point for that process. With better data, more rigorous evaluation and closer collaboration between energy and digital-infrastructure stakeholders, future regulatory designs can ensure that data centre expansion supports, rather than undermines, the long-term resilience and decarbonisation of energy network systems.

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